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Homework1 for NLP

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1.4 Questions

1.

is it true that every pickle with the chief of staff with every floor under every chief of staff under the sandwich in the sandwich under every president under a sandwich in a president under every pickle in the floor pickled every sandwich ?

is it true that a delicious sandwich pickled a floor with every perplexed sandwich ?

is it true that a president under the sandwich in every sandwich pickled a president under a pickle in every chief of staff ?

a perplexed chief of staff under the pickle on every floor under the delicious chief of staff in the floor in the perplexed floor in every president under a chief of staff in the president under every pickled floor in a chief of staff on the president under the president under every pickle understood a president on every perplexed sandwich in every chief of staff with every pickle on the pickle under a president under the pickle in every floor under every pickle in a pickled chief of staff with the pickle on the fine pickle in a chief of staff with a floor !

the perplexed pickle in the pickle kissed every floor on every pickle !

every sandwich understood every floor on the chief of staff in the president in a president .

is it true that a floor ate a delicious president with every president with the floor ?

a president on the fine floor with the delicious president under a president on every chief of staff on the sandwich under the floor on every floor with every fine floor on the pickle in the president with every sandwich in the chief of staff in a president under every sandwich under a president on the sandwich with every president under every sandwich under a president ate a pickle in every delicious sandwich under a sandwich under every president on a floor in a pickle in a floor in every sandwich with the pickled sandwich on the chief of staff on every president in every fine floor on a floor under the floor under a floor with a chief of staff under every president in every pickle in the sandwich with a pickle with every sandwich under the pickled delicious chief of staff in

every president under a pickled floor under a sandwich on the chief of staff under a delicious president in the sandwich on a chief of staff on a sandwich on every floor with the president under the floor on every president under the fine delicious sandwich under every pickle under a perplexed chief of staff under a president in the floor under every sandwich with the sandwich in every sandwich under the pickle in a president under a sandwich in a sandwich under the floor under the floor
in ... ... !

is it true that every chief of staff with a sandwich with the pickled sandwich under a president on a floor in the floor with the pickle with every sandwich in the pickle on a pickle on every president on a pickle kissed every sandwich on a pickle under the president in a sandwich in the chief of staff under the chief of staff with the pickle with the pickle under the chief of staff in every pickle under a chief of staff in a chief of staff in a president in every sandwich in the sandwich ?

the floor in every chief of staff pickled a chief of staff .

2.

(ROOT (S (NP (NP (NP (Det the)

(Noun president))

(PP (Prep on)

(NP (NP (NP (Det the)

(Noun president))

(PP (Prep on)

(NP (NP (Det a)

(Noun sandwich))

(PP (Prep under)

(NP (NP (NP (Det the)

(Noun chief

of

staff))

(PP (Prep on)

(NP (Det a)
(Noun floor))))
(PP (Prep under)
(NP (NP (Det every)
(Noun (Adj perplexed)
(Noun sandwich)))
(PP (Prep in)
(NP (NP (Det a)
(Noun (Adj fine)
(Noun president)))
(PP (Prep under)
(NP (Det a)
(Noun president)))))))))))
(PP (Prep under)
(NP (NP (NP (NP (NP (NP (Det the)
(Noun (Adj pickled)
(Noun pickle)))
(PP (Prep under)
(NP (Det every)
(Noun chief
of
staff))))
(PP (Prep under)
(NP (Det a)
(Noun floor))))
(PP (Prep in)

(NP (Det the)

(Noun pickle))))

(PP (Prep with)

(NP (NP (NP (NP (NP (NP (NP (Det every)

(Noun floor))

(PP (Prep with)

(NP (NP (Det a)

(Noun (Adj delicious)

(Noun chief

of

staff)))

(PP (Prep in)

(NP (NP (Det every)

(Noun floor))

(PP (Prep on)

(NP (Det a)

(Noun floor))))))))

(PP (Prep in)

(NP (Det a)

(Noun (Adj fine)

(Noun floor)))))

(PP (Prep with)

(NP (NP (Det every)

(Noun chief

of

staff))

(PP (Prep under)
(NP (NP (Det the)
(Noun sandwich))
(PP (Prep under)
(NP (Det a)
(Noun (Adj perplexed)
(Noun floor)))))))))
(PP (Prep with)
(NP (Det a)
(Noun (Adj perplexed)
(Noun president)))))
(PP (Prep under)
(NP (Det a)
(Noun floor))))
(PP (Prep under)
(NP (Det every)
(Noun chief
of
staff)))))))))
(PP (Prep on)
(NP (Det the)
(Noun sandwich))))
(VP (Verb ate)
(NP (Det the)
(Noun (Adj pickled)
(Noun president)))))

!)

(ROOT (S (NP (NP (Det every)
(Noun sandwich))
(PP (Prep on)
(NP (Det a)
(Noun pickle))))
(VP (Verb wanted)
(NP (Det a)
(Noun (Adj pickled)
(Noun floor)))))
!)

3.

a president ... .

every pickle ... !

is it true that the ... ... ?

a ... ...

is it true that a ... ... ?

every chief of staff ... !

every chief of staff ... !

... ...

... ... !

is it true that ... ... ?

(ROOT (S (NP (Det a)
(Noun chief

of

staff))

...)

.)

(ROOT (S (NP (Det a)

(Noun ...

...))

...)

!)

2.1 Questions

1. I think it's because some recursive rules used in the grammar tree.

Like NP->NP+PP and PP->Prep+NP

Because the probability of choosing NP->Det + Noun and NP->NP+PP is same, so if the program always choose NP->NP+PP it will recurse. And for PP, there's also an option to make NP->NP+PP, so this part will have a high chance to loop a lot of times.

2. Because the rule Noun->terminal appears 5 times, and Noun->Adj + Noun only appear 1 time, so the probability of Noun->terminal is 5/6, while adjectives is 1/6.

3. Increase the probability of NP->Det+Noun, from 1 to 2. Also the number 1 to 10 with Noun->Adj + Noun. So both these have 2/3 probability to be chosen.

4. 2 Verb ate

2 Verb wanted

0.5 Verb kissed

We could increase the probability of these words because ate and wanted can be used between either president or chief of staff and sandwich or pickle, while kissed just happens between president and chief of staff.

Also:

3 Prep with

1 Prep on

1 Prep under

0.5 Prep in

because every noun except floor can be followed by with, while under, on can just be used with floor, and in I think is not used properly here with those nouns.

5.

is it true that the president with the pickled pickled fine delicious fine delicious perplexed chief of staff with every chief of staff on the perplexed president with a delicious sandwich on a delicious president under every chief of staff ate a sandwich under every pickled chief of staff under every delicious floor ?

a president under every perplexed delicious pickled sandwich kissed every floor !

every delicious pickle on the perplexed fine fine pickled fine sandwich with the fine sandwich with every pickle ate a fine sandwich !

every floor pickled the pickled fine floor under a floor !

is it true that every pickled sandwich with a perplexed chief of staff wanted every chief of staff ?

a sandwich on a pickled pickled pickled delicious fine delicious delicious chief of staff kissed the fine pickled fine fine delicious pickled president with a pickled pickled president with every pickle !

is it true that the fine pickle with every delicious pickle pickled every pickle ?

a president ate every floor in every delicious fine pickled fine perplexed sandwich under the fine pickled delicious sandwich !

the fine delicious perplexed delicious pickled delicious perplexed pickled delicious pickle kissed a chief of staff .

is it true that every delicious perplexed pickled fine chief of staff pickled the fine pickled pickled perplexed pickled perplexed floor ?

2.3 Questions

9. For the basic grammar rules I separate verb into transitive verbs and intransitive verbs, add the conjunction “and” and adverb very, also add a special type of verb—like thought or suggested, after which we could use a clause sentence.

For the complete sentence rules, I add these two rules.

1 ROOT that S perplexed NP

1 ROOT it perplexed NP that S

For generalization, Verb_trans -> Verb_trans Conj Verb_trans , Noun->Noun conj Noun are also expanded. To generate adv like very very perplexed president… I use adv->adv recursion and add NP->det adv adj Noun, to use the prepositional phrase after an intransitive verb, I create the rule VP-> Verb_intrans PP.

10.

every very perplexed proposal thought that a delicious sandwich on every desk sighed in the floor .

it perplexed a delicious desk that the very fine proposal and the chief of staff and a very pickled floor and a very very fine president thought that Sally thought that the pickle worked in every very delicious floor .

every chief of staff thought that every very delicious president thought that every delicious sandwich wanted a fine floor on a desk and the fine floor with Sally and Sally and every very fine desk and every very delicious president .

it perplexed every delicious chief of staff that the desk thought that the pickled proposal worked .

the very fine chief of staff thought that every very perplexed chief of staff thought that a very very fine sandwich and the floor thought that the delicious floor thought that the very perplexed chief of staff wanted and understood and wanted and pickled and understood and understood and wanted and ate and ate and understood and wanted and ate and ate and wanted and wanted and pickled and ate and ate and pickled and kissed and understood and wanted and understood and ate and wanted and kissed and ate and understood and ate the delicious proposal !

it perplexed a floor that every very perplexed chief of staff thought that every proposal thought that a very very pickled floor thought that every perplexed pickle and every fine chief of staff with the president thought that every very delicious floor thought that the very fine desk thought that a proposal thought that the very very delicious sandwich worked .

is it true that a chief of staff wanted a floor ?

Sally worked !

a pickle thought that the very fine floor thought that a pickled desk on a pickle thought that a perplexed pickle with the proposal ate every sandwich .

the pickled proposal sighed with every pickle .

2.4 Remark

I think those tree-structured grammar rules are more or less hidden in the neural network. They trained the probabilities of the next words, but to some extent, to match the best probability, those words also should be structured in the human grammar, because that is the entire corpus of the human grammar. And the neural networks are also able to learn unnatural languages, because as long as someone said something in the Internet, they could use it as the training data, though the words someone said might not be natural. And this is also “the way those networks learn and grow”. They actually has much more languages data than us. So while we felt some point in languages unnatural, they may have already known how to deal with them, to which point they are “teachers”.

Because languages are spoken, generated from human, and it is a communication tool, with which people need to exchange information with each other, so in order to encode and decode information, languages need to have some common structure, so that people could easily understand each other. I think this is the main reason that languages still have those tree-structured rules. And as the complexity of the communication among people rose, as people no longer lived by group, as the modern globalization, people need much more precise and detailed communication, this is the reason they evolved the way they are.

I think grammars are not a good way to model linguistic data. Though the grammar rules are very natural and common to think of modeling, it is very complexed and hard to build a model that include all of the grammar rules. It would be quite huge and not possible to make. There are so many words, sentences, tenses, format… to follow. And sometimes people do use unnatural, or more “free” sentences. As far as I know, those the black English has some very different words and more modern or cooler, simpler words than the normal English. And in causal cases, the words are also different from the formal situation. Words can also have different meanings, or use as different functions. Some words can be either verb, noun, adjective…. So it is very hard to distinguish between such conditions when using the tree-grammar system as a predictor.

3.1 Questions

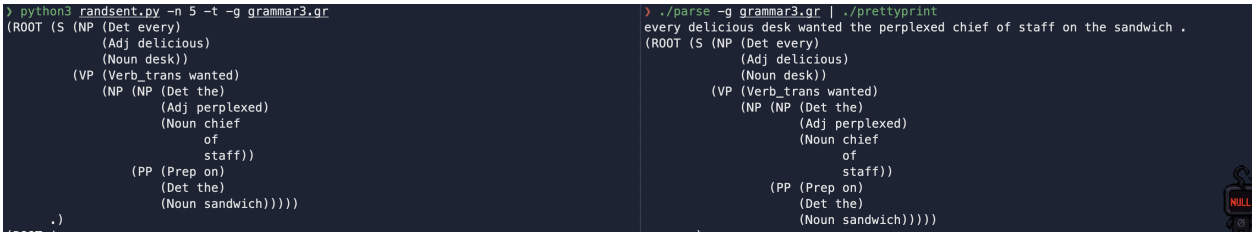
1. a. The other is :

```
(ROOT (S (NP (NP (NP (Det every)
                     (Noun sandwich))
                 (PP (Prep with)
                     (NP (Det a)
                         (Noun pickle))))
                 (PP (Prep on)
                     (NP (Det the)
                         (Noun floor))))
           (VP (Verb wanted)
               (NP (Det a)
                   (Noun president))))
       .)
```

b. The meaning of a sentence could change when the position of a preposition changes, for example, in this case, it means a pickle on the floor, while in a. It means every sandwich … on the floor. Depends on the words it decorates.

3.3 Questions Continued

2. I think in most cases like below, the randsent and the parser can parse the same



derivations.

Case 2: every very delicious floor thought that a very pickled chief of staff worked !

But here is an example that these two has some small differences:



A desk sighed with a floor !

The randsent.py treated with a floor as PP->Prep NP, NP->Det Noun,

```
every floor wanted a president under the floor with the chief of staff .
 (ROOT (S (NP (Det every) (Noun floor)) (VP (Verb wanted) (NP (NP (Det a) (Noun presi
dent)) (PP (Prep under) (NP (NP (Det the) (Noun floor)) (PP (Prep with) (NP (Det the)
 (Noun chief of staff)))))))) .)
# number of parses = 2
every floor wanted a president under the floor with the chief of staff in the sandwic
h .
 (ROOT (S (NP (Det every) (Noun floor)) (VP (Verb wanted) (NP (NP (NP (Det a) (Noun p
resident)) (PP (Prep under) (NP (Det the) (Noun floor)))) (PP (Prep with) (NP (NP (De
t the) (Noun chief of staff)) (PP (Prep in) (NP (Det the) (Noun sandwich)))))))) .)
# number of parses = 5
every floor wanted a president under the floor with the sandwich in the president wit
h the floor .
 (ROOT (S (NP (Det every) (Noun floor)) (VP (Verb wanted) (NP (NP (NP (NP (Det a) (No
un president)) (PP (Prep under) (NP (Det the) (Noun floor)))) (PP (Prep with) (NP (NP
 (Det the) (Noun sandwich)) (PP (Prep in) (NP (Det the) (Noun president)))))) (PP (Pr
ep with) (NP (Det the) (Noun floor)))))) .)
# number of parses = 14
```

While parser treated as PP->Prep Det Noun, which has ignored the middle part NP->Det Noun.

3.

I think there are five.

Those are the different derivations :

[every sandwich with ..][on the floor]…

[every sandwich][with a pickle on the floor]..

[every sandwich with ..][on the floor under …]

[every sandwich][with a pickle on the floor under…]

[every sandwich with a pickle on the floor][under …]

4.

I think it is because of the position of prepositions. For example, this sentence:

The president ate every chief of staff, this has only 1 parse.

```
is it true that every president kissed every pickle in the chief of staff un
der a president in the pickle ?
 (ROOT is it true that (S (NP (Det every) (Noun president)) (VP (Verb kissed
) (NP (NP (NP (Det every) (Noun pickle)) (PP (Prep in) (NP (Det the) (Noun c
hief of staff)))) (PP (Prep under) (NP (NP (Det a) (Noun president)) (PP (Pr
ep in) (NP (Det the) (Noun pickle)))))))) ?)
# number of parses = 5
```

But this sentence, because of the 3 continuous Preposition Phrases: in the … under … in the…, it has 5 different parses.

I did these experiments below.

It shows exactly the same as the number of brackets as the number of words ! (Words >= 3) So I think it is like, PP can be seen as a whole, and the number of PP plus the main body (which is one) those are like 3 words (if we have 2 PPs). So 3 words we have 2 brackets, 4 is 5, f(5) is 14, … So we can infer that for continuous PPs in a sentence, we count the number of PPs n , then n + 1 is seen as the whole words. For the number of parses, it follows the equation below.

$$f(n) = \sum_{k=1}^{n-1} f(k) \cdot f(n-k) \text{ for any } n > 1$$

5.

a.

Details are here: 1/3 (which is S . out of 3 S) * 1/2 (the probability of NP-> Det Noun) * 1/3 (“the” det prob) * 1/6(“president” prob, Noun->president) * 1/5 (Verb->ate) * 1/2(NP->Det Noun) * 1/3(Det->the) * 1/6(Noun->sandwich) which is 0.0000514 = 5.144 * e-5 !

Because the probability of generating the sentence is like the unigram, the prob of choosing these words is exactly the same as the certain parse here.

Because given the sentence, the sentence has only one parse. So is 1.

b.

Because given the sentence, it has 2 parses, so the prob of best_parse is 0.5.

c.

It is the -log(5.144e-05) - log(1.240e-09) = -43.833, sum of the -log probability of generating the above sentences.

d.

It is 2^2.435 =5.408 per word.

e.

It's failure and NaN. I think because there is no the president ate this sentence in the corpus, so there is actually no possibility to generate this sentence, so the program failed.

And P(best | sentence) = P(best, sentence) / P(sentence), since P(sentence) = 0(denominator), it is NaN.

6.

a.

I think it's 2.319 bits per word.

Command:

python randsent.py -g grammar2.gr -n 1000 | ./parse -P -g grammar2.gr

b.

grammar3.gr is 2.082 bits per word. It is more stable to generate the next word than grammar2.gr. I think because in grammar3.gr, it has several fix rules like PN->Sally, S->NP Verb_trans_clause that S, and Verb_trans_clause -> thought, adv->very, Conj->and, Verb_intrans->sighed, Verb_intrans->worked, so once these are chosen, the entropy is 0, because the probability of those words to be chosen is 1. So that the average entropy is somehow lower than the grammar2.gr.

c.

When it comes to grammar1.gr, the stand output said warning: … is not in grammar. Because I think the recursive rules: NP->NP PP and PP->Prep NP, so NP has much higher probability to stuck around then it exceeds the limit of expanding the non-terminals.

7.

It is true. In my program with 1000 sentences(22207 words corpus) generated from grammar2.gr, the grammar2 has 2.322 bits, while grammar3 has 2.962 bits and grammar.gr has 2.836 bits per word. I think it's because grammar3 has more rules than the grammar2, thus it does not easily choose the rules that inherited from grammar2 that could generate those grammar2 sentences. It has more extensions, so it's more "surprising" to generate the corpus from grammar2.

And grammar1 has somehow recursive rules, so it also made it difficult to generate the exact sentences from corpus of grammar2, though the rules are same from grammar2. And also because grammar2 has adjusted some weights to let the sentences generated much shorter, so the entropy of grammar1 is larger than that in grammar2.

4.1 Questions

I choose e and f to implement.

For the single VS plural agreement, I add one rule to S: S->PLNP Verb_plu NP, PLNP stands for all the plural nouns like the citizens, and NP and NP…, Verb_plu is a preterminal which contains the plural of a verb in present-tense. So only when S is divided into this case, the sentence will use plural verb. For different tenses, I divide the VP into 3 main tenses, which are VP_PRE, VP_PAST and VP_FUTR, then every tense has its detailed verb, like verb_trans_pre, which contains the simple, perfect, progressive and perfect + progr. And aslo verb_intrans_pre, verb_intrans_pre + PP…

I also adjusted some probabilities like Verb_intrans_pre Verb_intrans_pre Conj Verb_intrans_pre these style grammar rules to 2 because when I gave higher scores to this the program will generate much longer sentences. I raised adv ->very to 3, so that I could prevent the case : very very very very very very… which will make the sentence longer.

Here are some examples that my grammar generates:

is it true that every pickle on a proposal thought that every floor and Sally with every pickle eat Sally ?

that every proposal will be working with the floor perplexed every chief of staff .

the citizens eat every very perplexed chief of staff with a sandwich !

is it true that Sally kissed the very fine floor ?

that the citizens understand Sally with every sandwich perplexed every very perplexed chief of staff on the president .

it perplexed the perplexed sandwich that every very perplexed president thought that the very delicious chief of staff has worked in the president .

every pickled desk is sighing and has been sighing with the sandwich .

is it true that every pickled desk will choose the pickled president ?

that every pickled desk and a very very perplexed sandwich eat a very fine proposal perplexed the very fine president .

Sally has eaten the perplexed president .